HEREDITY (B)

GENETIC CROSSES

In minks, a single gene controls coat color. The allele for a brown (B) coat is dominant to the allele for silver-blue (b) coats.

1. A homozygous brown mink was crossed with a silverblue mink. There were 9 offspring in the F1 generation. What color were they?
2. Two of the offspring from the F1 generation above were mated. What would the ratio of brown to silverblue offspring be in the F2 generation?
3. What is the genotypic ratio of the F2 generation?

In domestic cats, the gene for Tabby stripes (T) is dominant over the gene for no stripes (t)

4. If a breeder crossed a heterozygous Tabby cat with a stripeless cat, what percent of the F1 generation would have tabby stripes?

OOmpas generally have gray faces, which is caused by a dominant gene. The recessive condition results in an orange face.

5. Two heterozygous OOmpahs are crossed. What proportion of the offspring will have orange faces?

Colorblindness is a sex-linked trait. A woman, whose mother is colorblind and whose father has normal vision marries a normal visioned man.

6. What is the woman's genotype?
7. What is the woman's phenotype?
8. What is the chance that her first offspring will be colorblind if male?
9. What is the chance that her first offspring will be colorblind if female?

GENETIC CROSSES (cont'd)

In humans, there are four types of blood; type A, type B, type AB, and type O.

10. What possible genotypes will produce B type blood?
11. What is the only genotype that will produce O type blood?
12. What is the only genotype that will produce AB type blood?

You are blood type O, and your partner is blood type AB.

13. Complete a Punnett square for the cross.
14. List the possible phenotypes of the offspring.

In the 1950's, a young woman sued film star/director Charlie Chaplin for parental support of her illegitimate child. Charlie Chaplin's blood type was already on record as type AB. The mother of the child had type A and her son had type O blood.

15. Complete the Punnett square for the cross between the mother of the child and Charlie Chaplin.
16. The judge ruled in favor of the mother and ordered Charlie Chaplin to pay child support costs of the child. Was the judge correct in his decision based on blood typing evidence? Explain why or why not. Refer to any Punnett squares to support your answer

HEREDITY (B)

Suppose a newborn baby was accidentally mixed up in the hospital. In an effort to determine the parents of the baby, the blood types of the baby and two sets of parents were determined. Baby 1 had type O Mrs. Brown had type B Mr. Brown had type AB. Mrs. Smith had type B Mr.Smith had type B.

17. Draw the Punnett squares for each couple.
18. To which parents does the baby belong? Refer to the Punnett squares in your justification.

A man with type B blood marries a woman with type A blood. They have the first child with blood type O.

19. What are the genotypes of the mother and the father?
20. Draw the Punnett square of the cross of the couple
21. What is the genotype and phenotype of the baby?

A cross between red-flowered snapdragons and white-flowered snapdragons produces offspring with pink flowers. Let R = red flowers and W = white flowers.

22. What is the genotype of a plant with a phenotype of red flowers?
23. What is the phenotype of a plant with a genotype of RW?

Suppose a pink-flowered plant is crossed with a pink-flowered plant.

24. Draw the Punnett Square.
25. What are the possible genotypes and phenotypes of the offspring?
26. What ratio of the offspring will have pink flowers?
27. What ratio of the offspring will have red flowers?

DNA REPLICATION AND STRUCTURE

28. DNA molecules are part of what class of macromolecule?
29. DNA molecules are made of monomers, termed _____?
30. The monomer is made of what parts?
31. DNA replication is crucial to the advancement of a cell in the cell cycle. In what stage does a cells **chromosomal** DNA go through replication?
32. What is the principal enzyme that is responsible for DNA replication?
33. DNA replication is directional. In what direction does it occur? How does this differ for the two strands?
34. What enzyme is responsible for unwinding the supercoil?
35. What enzyme is responsible for unzipping the double helix?
36. What is the name of the shape created after DNA is unzipped?
37. The leading strand has simple replication. The lagging strand has a more complicated set of reactions that replicate it. It is synthesized in pieces. What are these pieces called?
38. The template strand is 5'ATTGATCCTAGTCA3'. What will the replicated strand be?

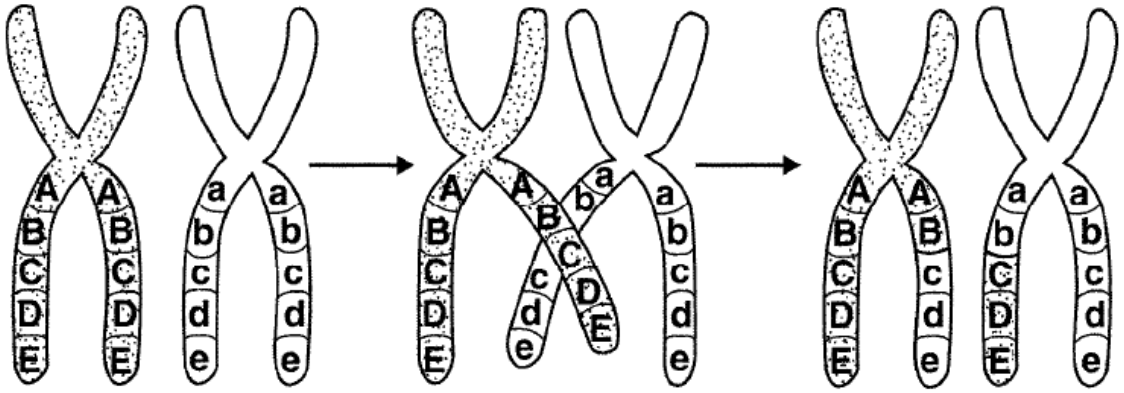
MITOSIS AND MEIOSIS

39. Mitosis occurs in what kinds of cells?
40. Mitosis is split into many stages. Name the 5 major stages (exclude prometaphase in all answers).
41. How many times does DNA replicate during the cell cycle and mitosis?

HEREDITY (B)

42. How many times do the cells divide in mitosis?
43. What is the ploidy of a human's parent cell?
44. What will the ploidy be of the daughter cells?
45. What is the main purpose of mitosis?
46. Which of the following happens when a cell divides?
    a) The cell's volume increases.
    b) It becomes more difficult for the cell to get enough oxygen and nutrients.
    c) The cell has DNA overload.
    d) Each daughter cell receives its own copy of the parent cell's DNA.

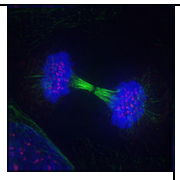
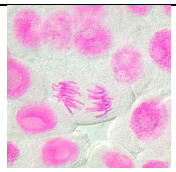
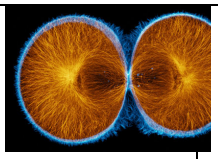
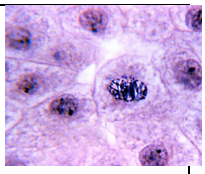
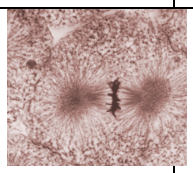
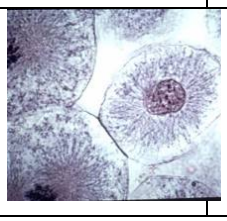
47. Which pair is correct?
    a) G1 phase, DNA replication
    b) G2 phase, preparation for mitosis
    c) S phase, cell division
    d) M phase, cell growth
48. When during the cell cycle is a cell's DNA replicated?
    a) G1 phase
    b) G2 phase
    c) S phase
    d) M phase
49. Which event occurs during interphase?
    a) The cell grows.
    b) Centrioles appear.
    c) Spindle fibers begin to form.
    d) Centromeres divide.



50. (**Refer to previous diagram**) What is shown in the figure?
    a) Independent assortment
    b) Anaphase I of meiosis
    c) Crossing-over
    d) Replication

HEREDITY (B)

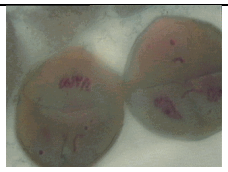
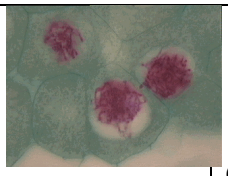
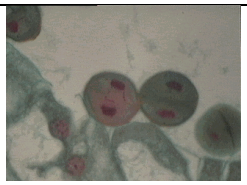
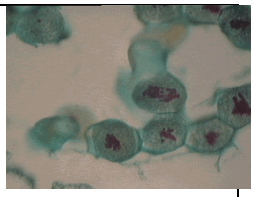
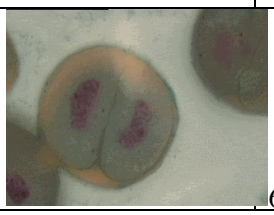
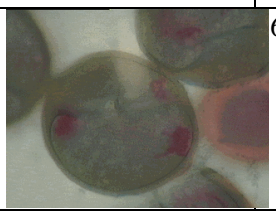
For the following images, write down the stage of mitosis the cell is in.

51. 	52. 	53. 	54. 
55. 	56. 	57. During what phases are there chromatids?	58. During what phases are there chromosomes ?

59. **Short Answer:** Distinguish and Explain the following terms briefly
    a) Chromatin
    b) Chromosomes
    c) Chromatid
    d) Sister vs nonsister chromatid
    e) Homologous Chromosomes
    f) Asters
    g) Telophase vs Cytokinesis
    h) Histone proteins

For the following images, state what phase of meiosis the cell is in. *

*Questions 60, 64, 66, 67 will not count for credit.

60. 	61. 	62. 	63. 
64. 	65. 	66. 	67. List which question numbers have images from meiosis II.

68. Explain the following terms briefly
    a) Meiosis I vs II
    b) Ploidy before and after
    c) Meiosis purpose and Mitosis purpose

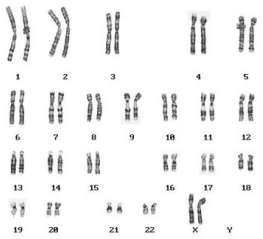
HEREDITY (B)

For Questions 51 and 52, write the gender on your answer sheet. For Questions 53-58, label the nondisjunction disorder on your answer sheet.

69.



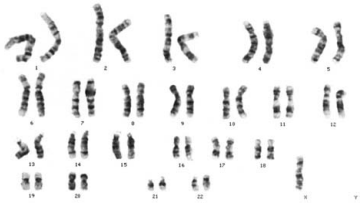
70.



71.



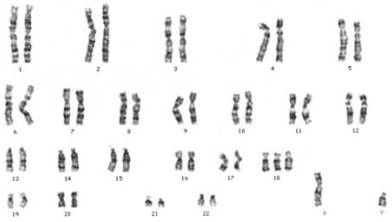
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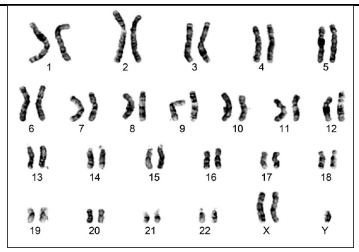
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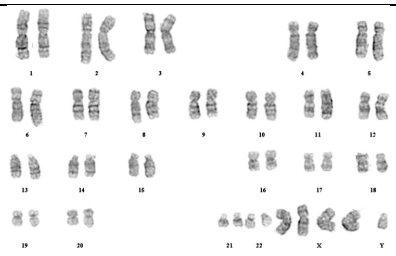
74.



75.



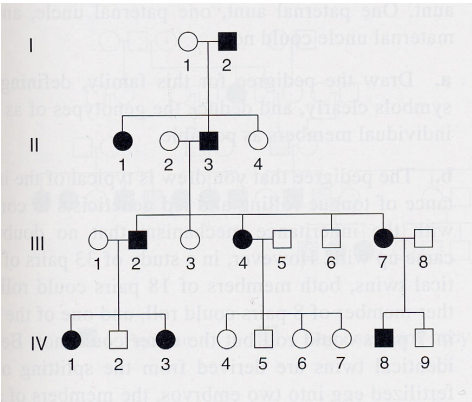
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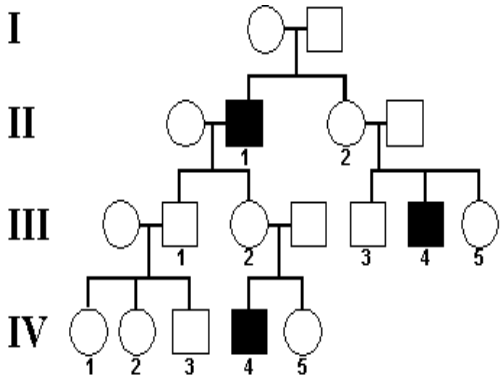


~~77. What is the apparent gender of the karyotype in Question 34?~~ Question redacted

HEREDITY (B)

Pedigree Analysis





78. Referring to the above pedigree (left), the inheritance of the disease by II-3 rules out what type of inheritance? Why?
79. Referring to the above pedigree (right), what is the type of inheritance imaged?

Essay Questions (tie breakers)

62. Discuss Mendel's laws of segregation and independent assortment. Explain how the events of meiosis I account for the observations that led Mendel to formulate these laws. (10 pts)

63. Meiosis reduces chromosome number and rearranges genetic information. (10 pts)
    a) Explain how the reduction and rearrangement are accomplished in meiosis.
    b) Several human disorders occur as a result of defects in the meiotic process. Identify ONE such chromosomal abnormality; what effects does it have on the phenotype of people with the disorder? Describe how this abnormality could result from a defect in meiosis.

HEREDITY (B)

reesabp 's 2014 User-Submitted Test - Repaired practice edition

Division B | Heredity | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

q67. (6 points) Authored replacement for source item 67: Solve this case and explain your reasoning: A diploid cell with 2n = 8 undergoes meiosis. How many chromosomes and chromatids are in each cell at the end of meiosis I and at the end of meiosis II?

Answer: _____
_____
_____

repair-case. (4 points) Authored written substitute for unavailable source material: Name an experimental question that PCR could answer.

Unit: PCR and Sanger sequencing.
Worked-case data: A PCR starts with 50 copies of a target. Assuming perfect doubling, how many copies are present after 10 cycles?
Separate investigation observations:
Positive control: Known target DNA gives a 450 bp band.
Negative control: Water gives no band.
Sample 1: One band at 450 bp.
Sample 2: No band, but a later test shows its DNA is intact.

Answer: _____
_____
_____